



INGB413 EXAM ASSIGNMENT

INGB413

QUALITY ASSURANCE

INDIVIDUAL EXAM ASSIGNMENT

FIRST OPPORTUNITY

22 MAY 2026

TOTAL: 100



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Assignment Objective

The objective of this assignment is for you to demonstrate that you have mastered a fundamental understanding of quality management principles. Furthermore, this assignment assesses whether you have achieved the Engineering Council of South Africa's (ECSA) Graduate Attribute (GA) 4 at Exit Level.

ECSA Graduate Attribute 4: Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

General Instructions

For this individual assignment, you will have the opportunity to gather data during an experiment in the Learning Factory on **22 May 2026**. After the data gathering, you need to submit a report on e Fundi (refer to instructions below) by **30 May 2026 @ 09:00**. This report will be assessed to determine whether you have successfully completed the assignment and achieved ECSA GA 4 at Exit Level.

Practical arrangements

The time slot for you to conduct your experiment in the Learning Factory is **Friday, 22 May 2026 @ 09:30**. Data collection can only be done during this allocated time slot.

- Please arrive **10 minutes prior** to your assigned time and wait outside the Learning Factory.
- Bring along a pen and paper to record the results (refer to the technology-free zone information stipulated below).
- You are allowed to print this assignment and bring it along with you to refer to during data gathering.

Occupational Health and Safety in the Learning Factory

The Learning Factory is a simulated factory environment, and therefore, relevant health and safety protocols should be adhered to.

The Learning Factory is a technology-free zone. No electronic devices (such as phones/tablets/laptops) will be allowed. There are lockers in the Learning Factory where you can lock up your personal belongings and keep the key with you for the duration of the practical experiment.

- Please wear closed shoes when entering the facility.
- Each student will be provided with a safety vest associated with the production line to which you are assigned. Please wear this safety vest at all times.
- Please pay attention to all the safety instructions conveyed to you by the Learning Factory safety officer at the start of the designated session. This information will include details on safety exits and protocol to follow in case of an emergency or injury.



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Data Gathering Experiment

A tangram puzzle is a traditional Chinese puzzle that consists of seven geometric shapes that can be rearranged to form a variety of new shapes and figurines.

Tangra Inc. is a tangram manufacturing company that sources puzzle pieces from different suppliers and ships them to customers. *Tangra Inc.* also supplies customers with pictures of the different figurines that can be built from the tangram pieces. Customers have recently complained that when they build the figurines from *Tangra Inc.* tangram pieces, the figurines cannot be built to look the same as the figurines supplied by *Tangra Inc.* This is due to some of the tangram pieces that are defective.

As the appointed industrial engineer responsible for the quality assurance of *Tangra Inc.*, you have consulted with the marketing team, who emphasise the importance of maintaining *Tangra Inc.*'s reputation of being an innovative company that supplies high-quality products. You have therefore decided to assemble and ship each tangram already assembled in the format of a figurine to prove to customers that the set they receive does not have any defective parts.

However, given the current situation with defective parts received from suppliers, you need to investigate whether enough tangrams can be assembled as figurines to keep up with current demand, whilst also identifying defective pieces efficiently. You also need to determine whether the process will be able to run at a 2-sigma level (in terms of defective parts per million).

You conduct two quality tests based on two different sampling processes.

Process 1

- a. Receive raw material in bulk
- b. Take samples of 5 and test for defects (sample size (n=5))
- c. Remove all defective parts
- d. Assemble figurine

Test process 1 by following these process steps and assemble as many figurines of one specific type as possible in 20 minutes. Record all the defects, the number of completed figurines as well as any other relevant information.

Process 2

- a. Receive raw material already sorted by colour in lin bins at the assembly station
- b. Pick parts from the lin bins
- c. Start assembling a figurine
- d. If you find any defects, discard the entire assembly
- e. Count the number of puzzle pieces already placed and the number of defective pieces in each discarded assembly

Test process 2 by following these process steps and assemble as many figurines of one specific type as possible in 20 minutes. Record all the defects, the number of completed figurines, as well as any other relevant information.

Cost information

The following information regarding the cost of raw materials and defects is known and shown in table 1:

Table 1 Tangra Inc costing information

Item	Cost
Puzzle piece	R2.00
Rework per puzzle piece	R1.00
Defective semi-assembled item	R20.00

Each completed product can be sold for **R25.00**.

Report instructions

Write a report for the management team of *Tangra Inc.* where you highlight how you have investigated the data and what the findings of the two processes in terms of quality parameters are. Clearly explain how the experimental approach followed is suitable to investigate the quality parameters and be sure to refer to clarify all assumptions and limitations. Make use of quality assurance principles (which include statistical process control techniques) and provide insights on which of the two processes results in better quality outcomes and what the effect on profitability is. The process steps of both processes need to be clearly communicated to the management team.

Ensure that you support the use of quality management and assurance techniques with relevant research on aspects related to both quality from the customer's perspective and the organisation's perspective. This implies that you need to do research on quality assurance principles as well as specific techniques used. Clearly state all assumptions. Also, be sure to test relevant hypotheses, include valid control charts and use a relevant measure to comment on the capability of the process.

This is an individual assignment, and you are not permitted to work on the reports and data analysis with anyone else

Your report needs to be typed, and no handwritten or scanned reports will be accepted. Your report needs to consist of at least the following headings, but you are free to structure the report as you see fit.



- i. Background
- ii. Data collection and analysis
- iii. Literature review
- iv. Conclusion
- v. References

There is no strict limit on the number of pages allowed for the report, but a general guideline is no more than 20 pages.

Submission information

Submit the report as a PDF document and your data separately as an Excel document on eFundi under the GA Assignment by **Saturday 30 May 2026 @ 09:00**

Name your report as follows:

Name_student number_INGB413GA for example

Maria van Zyl_12345678_INGB413GA

Only individual assignments submitted via eFundi will be considered and no reports that are e-mailed will be accepted.

Important Information

When you submit your assignment, a Turnitin report will be generated that will be reviewed as part of your assessment. There is no minimum percentage of similarity required but the examiners will look for plagiarism issues in this report.

1. **WARNING:** Plagiarism of any kind is a serious offence and will result in severe steps in accordance with NWU policy.
2. **PLAGIARISM INCLUDES SELF-PLAGIARISM AND THIS IMPLIES THAT NO WORK CAN BE SUBMITTED FOR THIS ASSIGNMENT THAT WAS SUBMITTED FOR A PREVIOUS ASSIGNMENT. YOU THEREFORE CAN NOT SUBMIT ANY PARTS OF ANOTHER ASSIGNMENT FOR THIS ASSIGNMENT.**
3. Although it is permitted to make use of writing assistance tools such as Grammarly, you may not use large language AI models (i.e. ChatGPT) to write the report or analyse data.
4. Reports will be checked for the unethical use of AI
5. You can only pass this module if you pass the GA assignment (above 50%) and your participation mark + GA Assignment mark is 50% or more.
6. **No late assignments will be accepted for any reason.**
7. **No assignments will be accepted via e-mail**
8. **If you do not submit on eFundi on time you will fail the GA Assignment.**